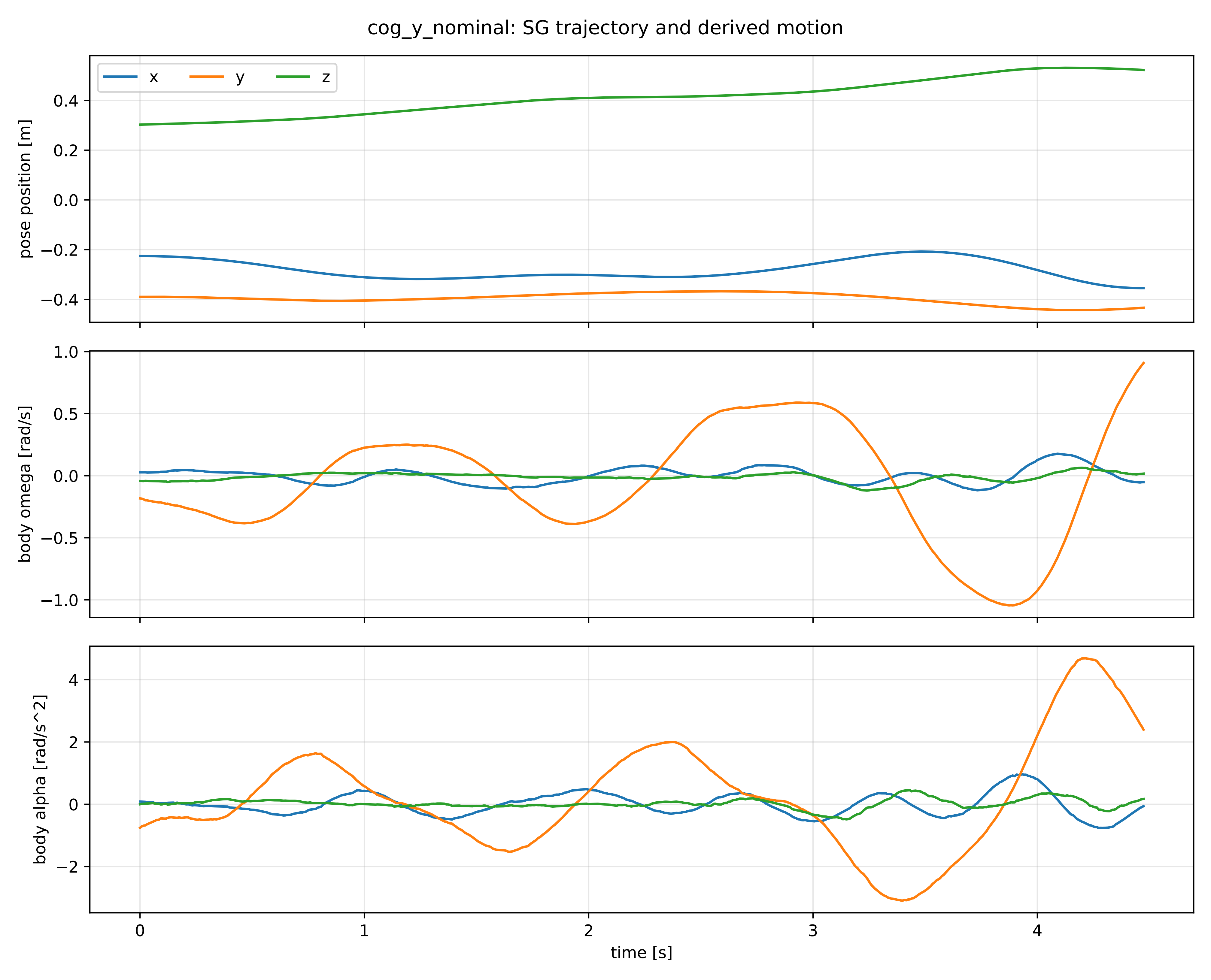
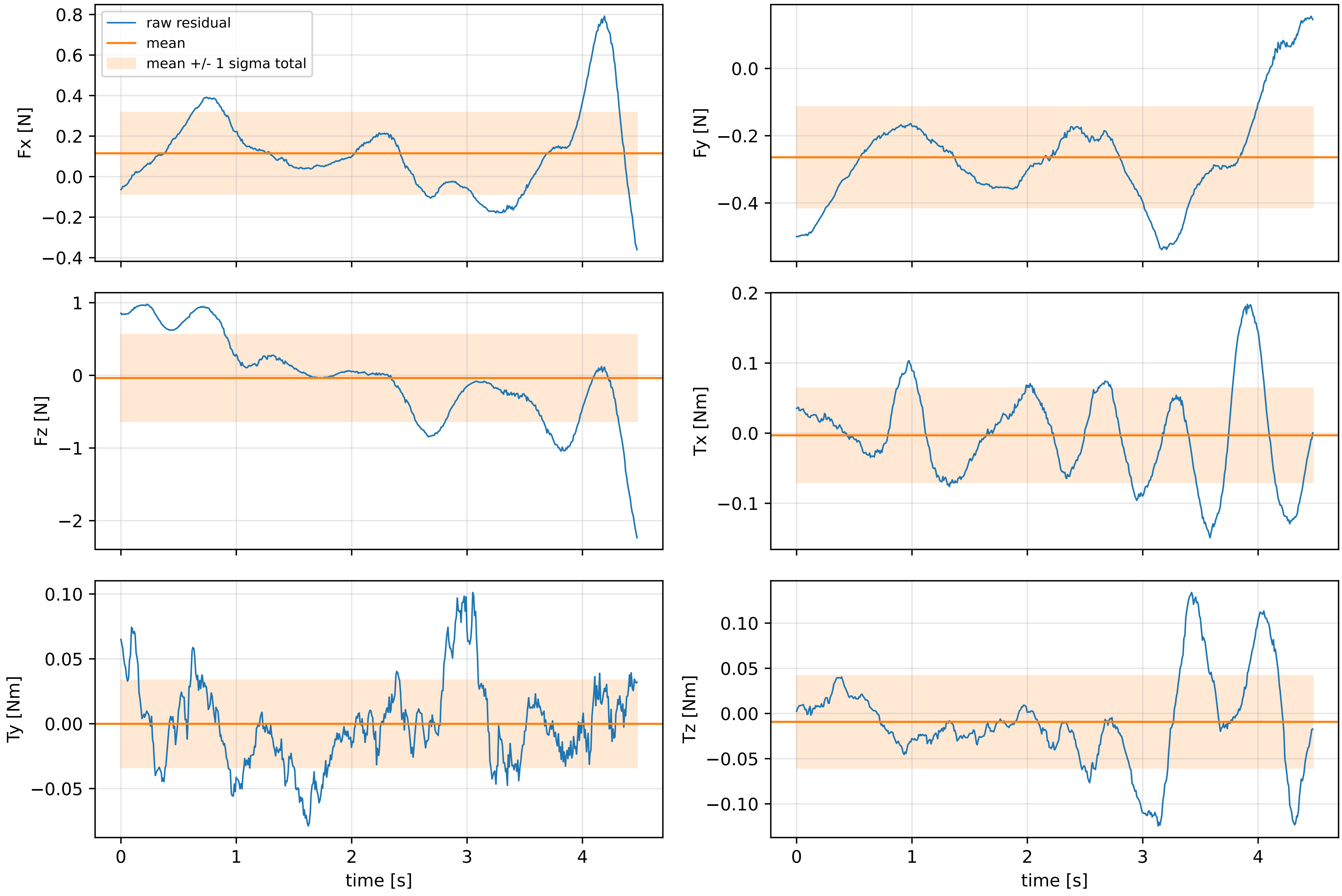


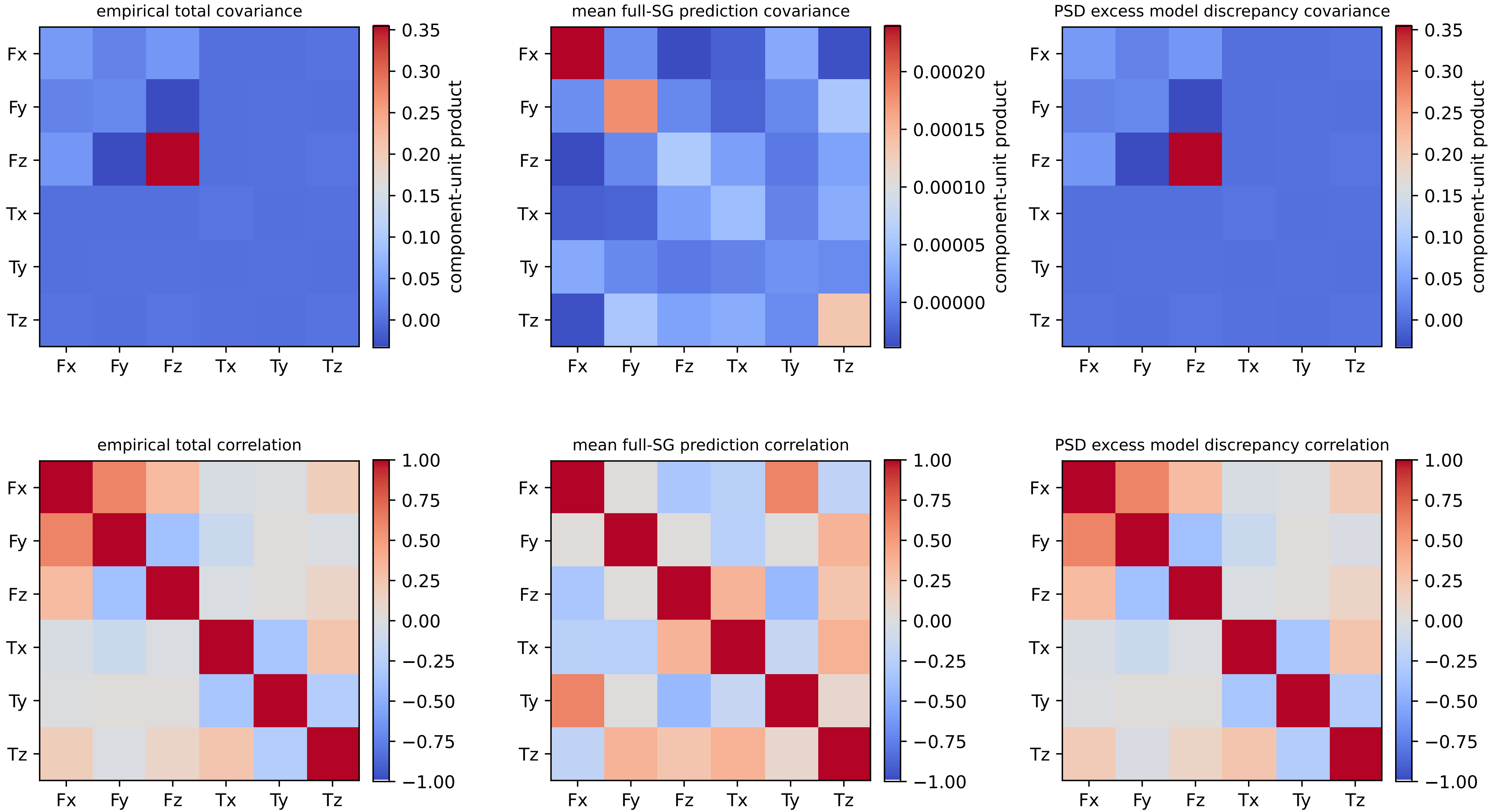
cog_y_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]

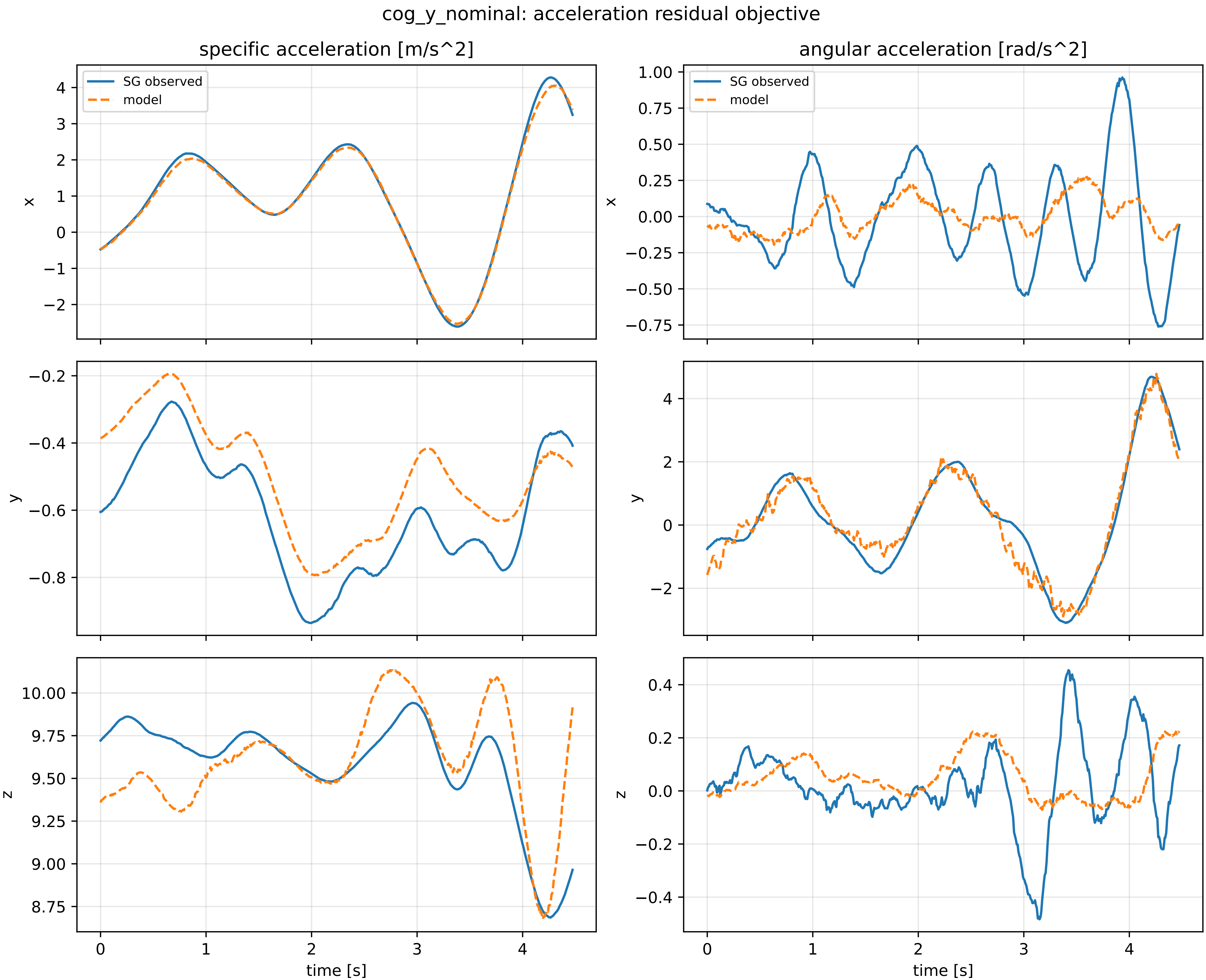
cog_y_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



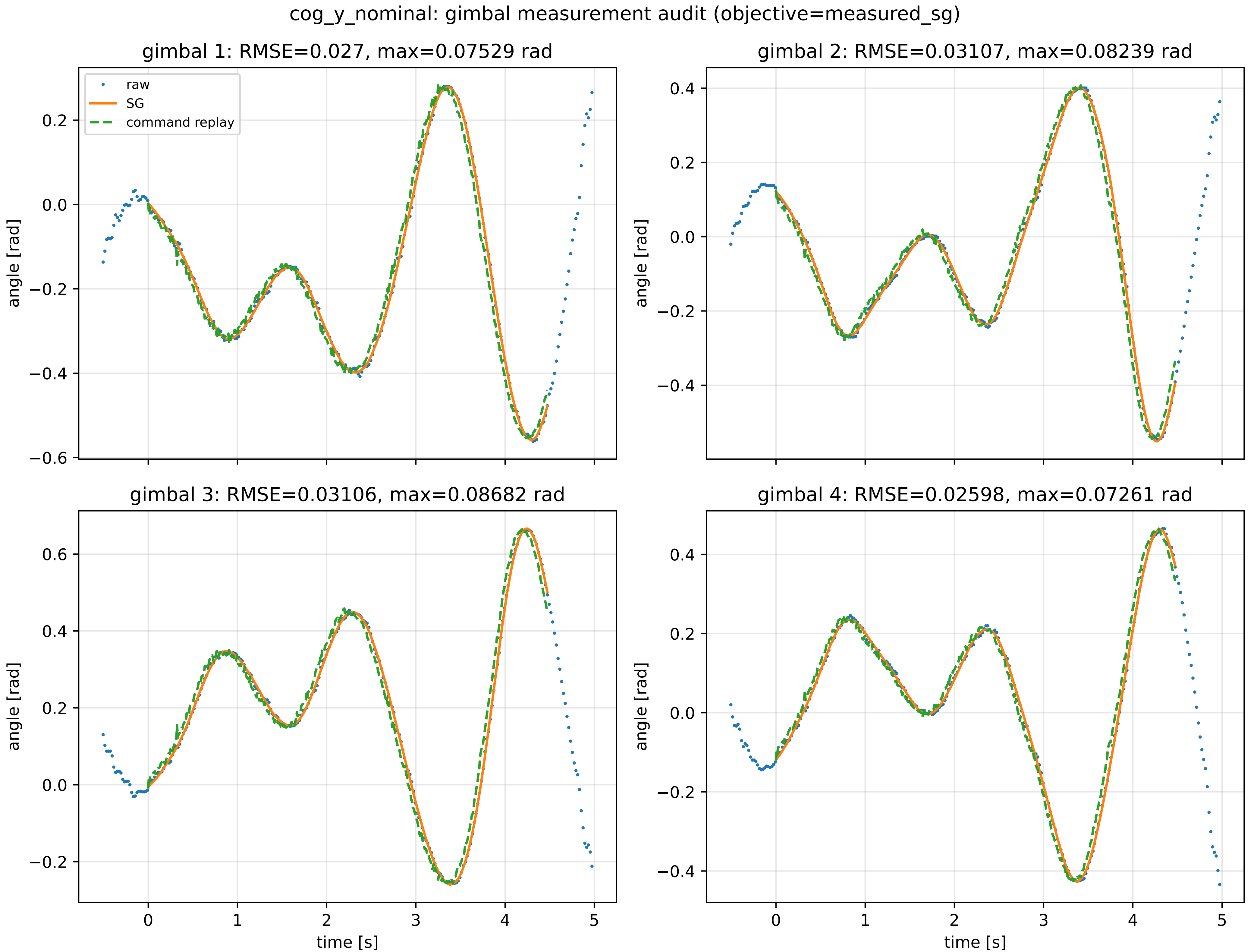
vector RMS about zero: force 0.7075 N, torque 0.090941 Nm; component std about mean: force [0.2004 0.1495 0.5954], torque [0.0669 0.0335 0.0508]

cog_y_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.004 0.006 0.05 0.362]

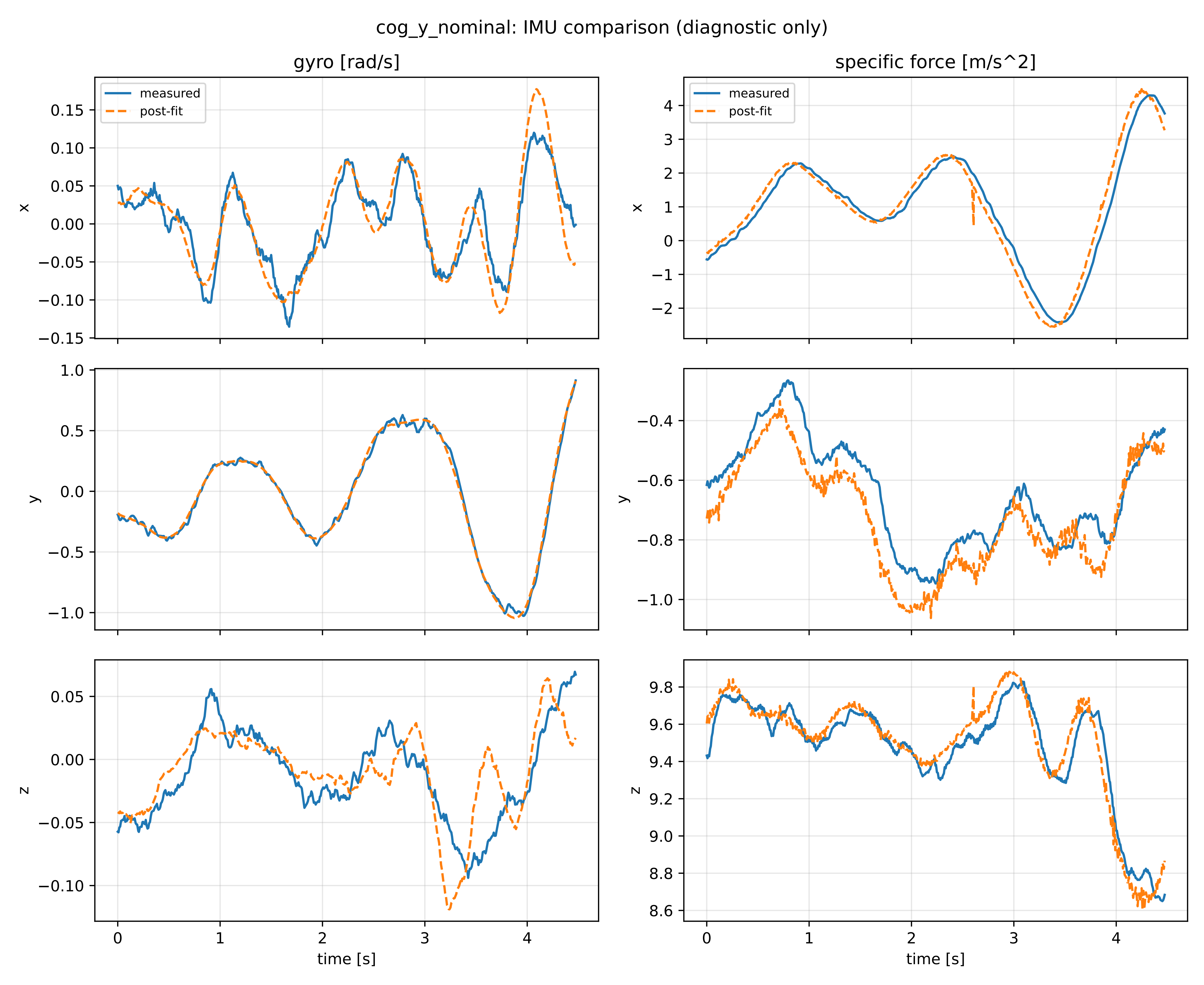




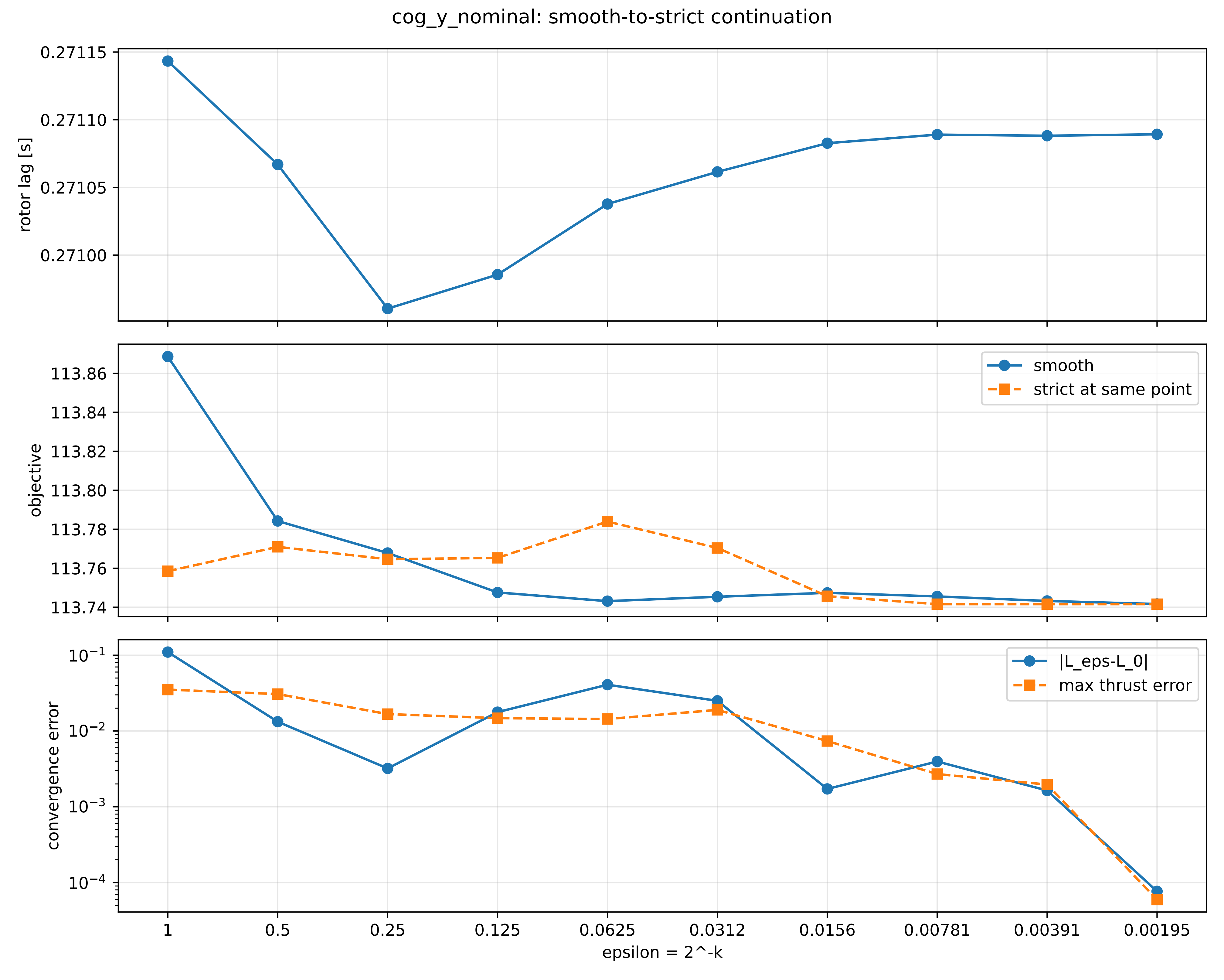
cog_y_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



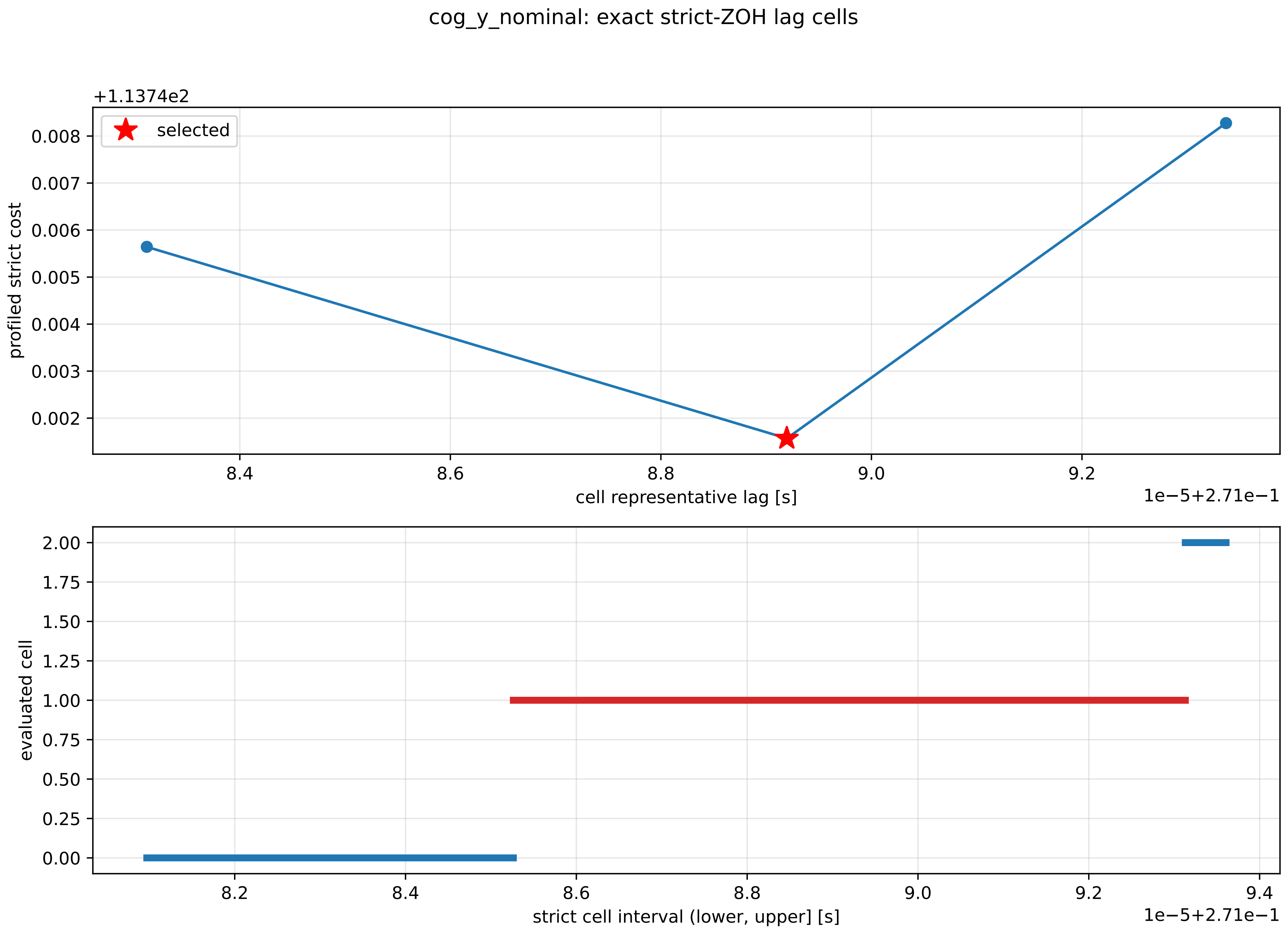
cog_y_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.027, max=0.07529 rad
gimbal 2: RMSE=0.03107, max=0.08239 rad
gimbal 3: RMSE=0.03106, max=0.08682 rad
gimbal 4: RMSE=0.02598, max=0.07261 rad
raw
SG
command replay
angle [rad]
time [s]



cog_y_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



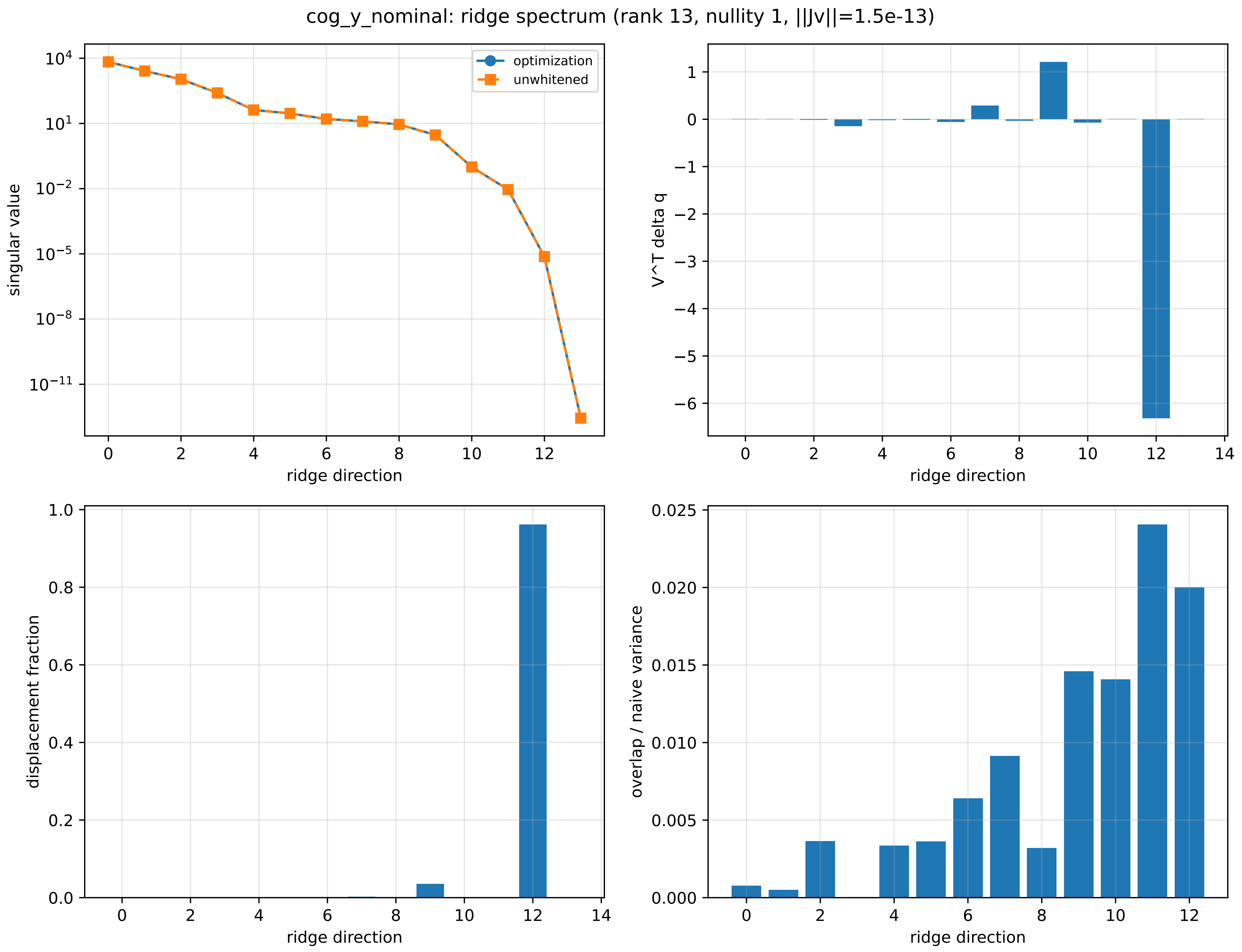
cog_y_nominal: smooth-to-strict continuation
rotor lag [s]
0.27115
0.27110
0.27105
0.27100
objective
113.86
113.84
113.82
113.80
113.78
113.76
113.74
smooth
strict at same point
convergence error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k



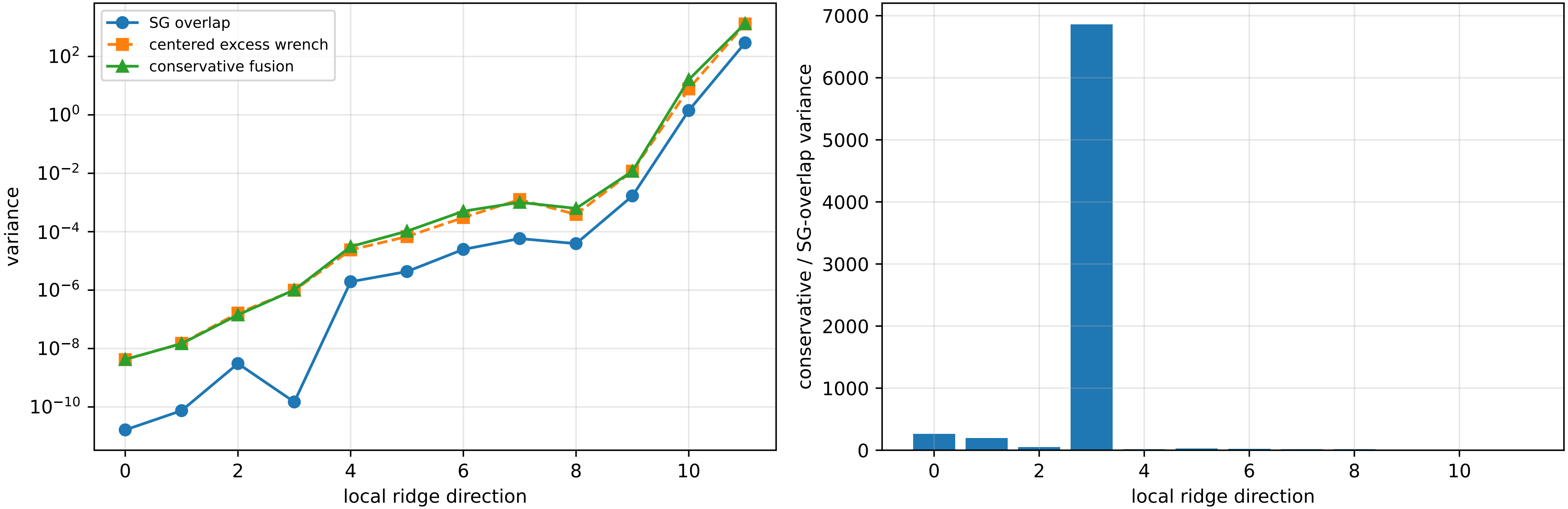
selected (0.271085262, 0.27109313] s; final smooth 0.271089244 s; data boundary=False

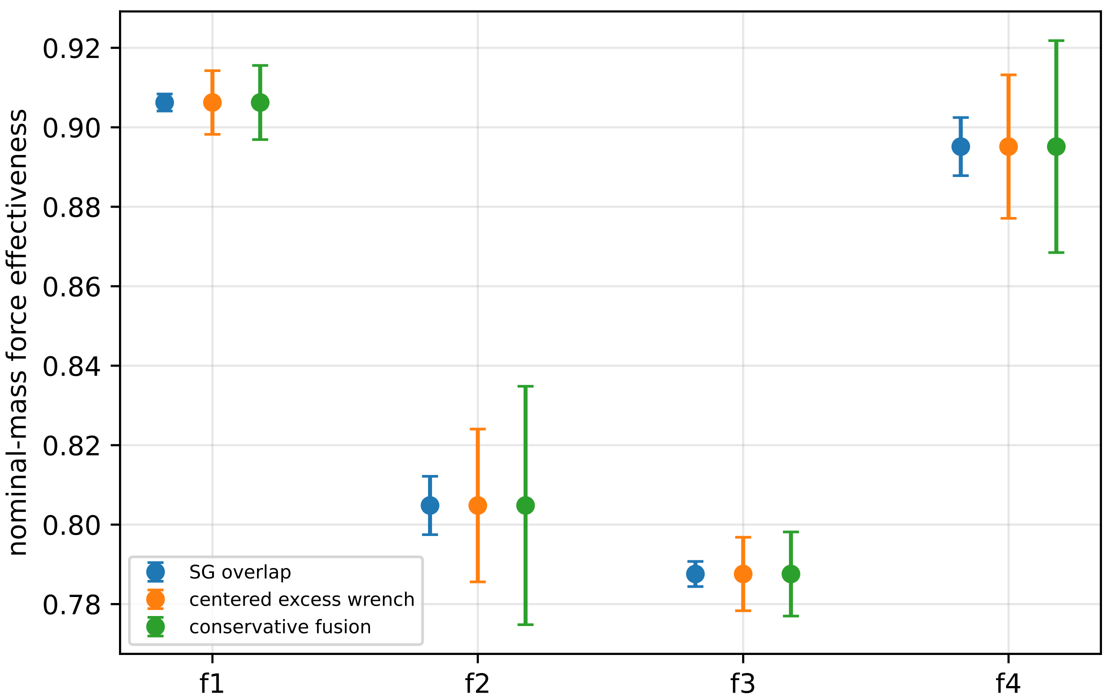
cog_y_nominal: parameter estimate

```
Case: cog_y_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.91351349
inertia [kg m^2]:
[[ 4.29743445e-01  1.18515608e-02  5.34963297e-04]
 [ 1.18515608e-02  1.63197070e-01 -9.59375651e-03]
 [ 5.34963297e-04 -9.59375651e-03  5.92508260e-01]]
CoG body [m]: [-1.82032651e-02 -3.05103942e-05  1.10204136e-02]
force effectiveness: [1.89353175 1.6816196  1.64558789 1.87034416]

scale-free J/m [m^2]:
[[ 8.74615377e-02  2.41203384e-03  1.08875919e-04]
 [ 2.41203384e-03  3.32139254e-02 -1.95252471e-03]
 [ 1.08875919e-04 -1.95252471e-03  1.20587490e-01]]
scale-free f/m: [0.38537225 0.34224381 0.33491063 0.3806531 ]

rotor lag cell: (0.271085262, 0.27109313] s
representative: 0.271089196 s
```


cog_y_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.5e-13)
singular value
ridge direction
optimization
unwhitened
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction

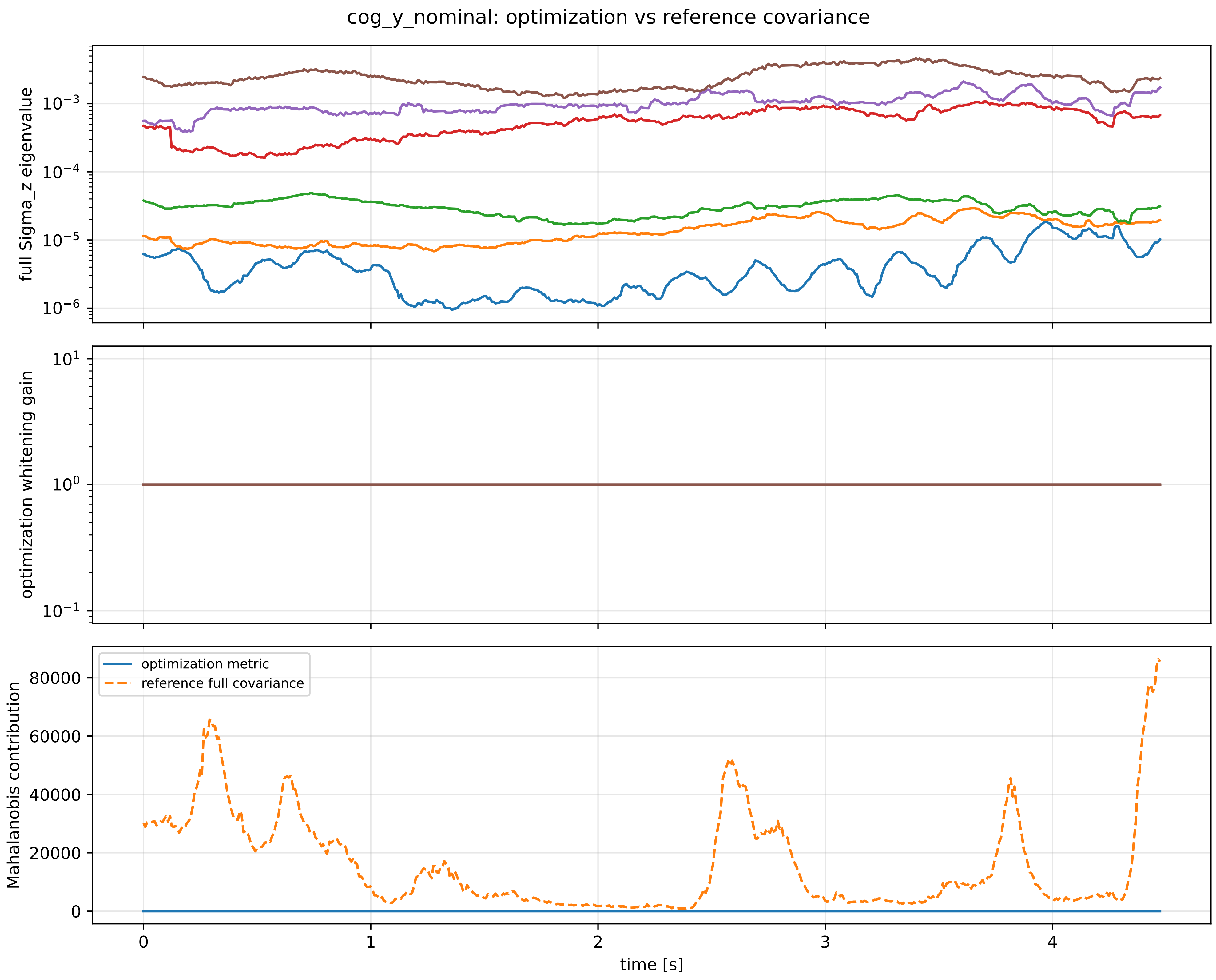
cog_y_nominal: Conservative fusion uncertainty



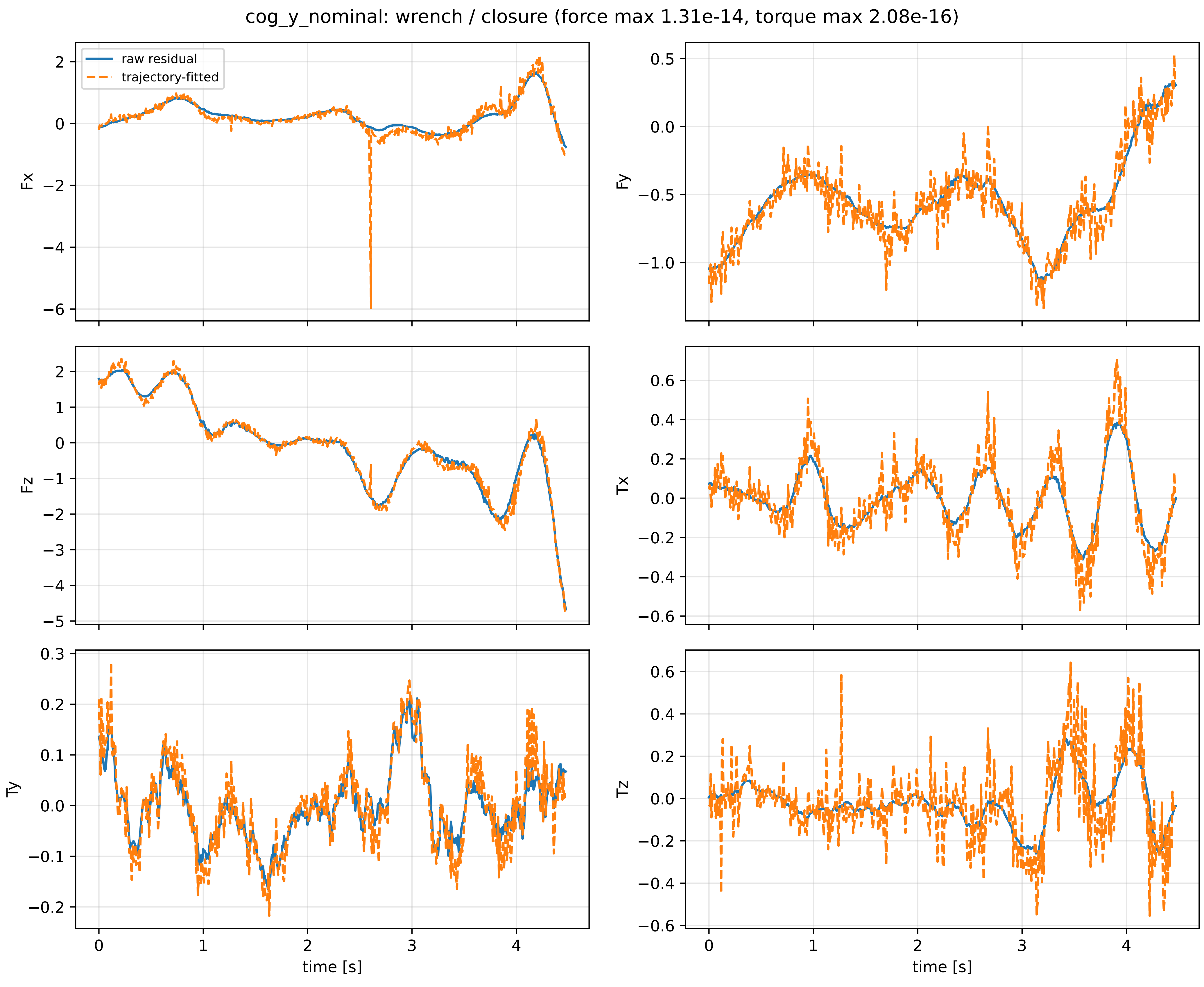



```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 232.1
wrench-to-acceleration recovery max error = 2.66e-15
exact scale gauge ridge index = 13 (alignment 1)

index 11: variance=1345, ratio=4.587
  log_mass=-0.000367; second_moment_diag_1=-0.0203
  second_moment_diag_2=+0.0113; second_moment_diag_3=+0.0108
  second_moment_offdiag_12=+0.229; second_moment_offdiag_13=-0.973
  second_moment_offdiag_23=-0.0329; cog_x=-1.54e-05
  cog_y=-9.2e-07; cog_z=-8.31e-07
  log_force_eff_1=-0.000308; log_force_eff_2=-0.000443
  log_force_eff_3=-0.000441; log_force_eff_4=-0.0003

index 10: variance=16.09, ratio=11.41
  log_mass=+0.00701; second_moment_diag_1=+0.00787
  second_moment_diag_2=+0.301; second_moment_diag_3=-0.344
  second_moment_offdiag_12=-0.0194; second_moment_offdiag_13=+0.0251
  second_moment_offdiag_23=-0.889; cog_x=+0.000602
  cog_y=-0.000714; cog_z=+2.95e-06
  log_force_eff_1=+0.00714; log_force_eff_2=+0.0137
  log_force_eff_3=+0.00618; log_force_eff_4=+0.00197

index 9: variance=0.01177, ratio=7.008
  log_mass=-0.151; second_moment_diag_1=-0.151
  second_moment_diag_2=+0.88; second_moment_diag_3=+0.0413
  second_moment_offdiag_12=-0.0128; second_moment_offdiag_13=+0.00187
  second_moment_offdiag_23=+0.274; cog_x=-0.00902
  cog_y=-0.000906; cog_z=-0.000507
  log_force_eff_1=-0.112; log_force_eff_2=-0.19
  log_force_eff_3=-0.201; log_force_eff_4=-0.117
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.



cog_y_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-3}
10^{-4}
10^{-5}
10^{-6}
optimization whitening gain
10^{1}
10^{0}
10^{-1}
Mahalanobis contribution
optimization metric
reference full covariance
80000
60000
40000
20000
0
0
1
2
3
4
time [s]



cog_y_nominal: wrench / closure (force max 1.31e-14, torque max 2.08e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

cog_y_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_cog_y_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/cog_y_nominal.j
SHA256: 3efc6d36ba4fff0dc12548d07dd00c0a5c4bcb02e99decb13ffa2adae4ce3f12

data objective:  113.741568039
prior objective: 1.30972815202e-06
total objective: 113.741569348

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: cog_y_nominal (cog_position_body_m)
  components: y
  target: [-3.05265789e-05]
  std:    [1.e-05]
  final:  [-3.05103942e-05]
  error:  [1.61847345e-08]
  standardized residual: [0.00161847]
  factor objective: 1.30972815202e-06
```